Security threats	Facilitating the conduct of cyber attacks, weapon development, and security breaches	Generating code to hack into government systems (Burgess, 2023; Shevlane et al., 2023) ±
Human Autonomy & Integrity Harms		
Violation of personal integrity	Non-consensual use of one's personal identity or likeness for unauthorised purposes (e.g. commercial purposes)	Generating a deepfake image, video, or audio of someone without their consent (Hunter, 2023)*
Persuasion and manipulation	Exploiting user trust, or nudging or coercing them into performing certain actions against their will (c.f. Burtell and Woodside (2023); Kenton et al. (2021))	A personalised AI assistant persuading someone to harm themselves (Xiang, 2023)*
Overreliance	Causing people to become emotionally or materially dependent on the model	Skill atrophy (e.g. decreased critical thinking skills) from excessive model use (Bai et al., 2023b) ±
Misappropriation and exploitation	Appropriating, using, or reproducing content or data, including from minority groups, in an insensitive way, or without consent or fair compensation	Training an image-generating model on an artist's work without their consent (Chen, 2023)*
Socioeconomic & Environmental Harms		
Unfair distribution of benefits from model access	Unfairly allocating or withholding benefits from certain groups due to hardware, software, or skills constraints or deployment contexts (e.g. geographic region, internet speed, devices)	Better hiring and promotion pathways for people with access to generative AI models (Gmyrek et al., 2023) ±
Environmental damage	Creating negative environmental impacts though model development and deployment	Increase in net carbon emissions from widespread model use (Patterson et al., 2021) ±
Inequality and precarity	Amplifying social and economic inequality, or precarious or low-quality work	Lower pay and precarious conditions for creative professionals (e.g. illustrators or sound designers) (Zhou, 2023)*
Undermine creative economies	Substituting original works with synthetic ones, hindering human innovation and creativity	AI-generated artefacts leading to a homogenisation of aesthetic styles (Epstein et al., 2023) ±
Exploitative data sourcing and enrichment	Perpetuating exploitative labour practices to build AI systems (sourcing, user testing)	Exposing human annotators to toxic audiovisual content (Perrigo, 2023)*

References marked with (*) indicate real-world examples; those marked with (±) are hypothetical and indicate anticipated risks.

A.2. Evaluation methods per layer

A.2.1. Capabilities layer

Human annotation

While automated annotation is increasingly common, it is usually calibrated against *human annotation*. Human annotation is taken as the ground truth for most risk areas[11] because whether a given AI system output is offensive, misleading, or indicates other risks of harm is subject to human judgement. Human annotators are shown a series of model outputs and tasked to judge these against a given set of criteria, such as a pre-existing set of rules or content policies (e.g. Glaese et al. (2022); Thoppilan et al. (2022)). For example, human annotators can leverage expertise to assess whether a given model output is likely to polarise political debates or sow division between social groups, although targeted expert annotation is relatively rare. Different annotators may provide different types of domain expertise, such as medical expertise or different lived experiences (Abercrombie and Rieser, 2022). To assess social and ethical harms, human annotators can assess whether a given AI system output indicates risks of downstream harm – for example, by showing factually incorrect, offensive, or misleading content.

Human annotation presents a number of limitations. Depending on the annotation task, annotators may be exposed to risks of harm (Stoev et al., 2023). Human annotators are often employed under precarious conditions (Gray and Suri, 2019). While annotation that leverages specific forms of human expertise is valuable, it can be difficult to source via commonly used annotation platforms (Zhang et al., 2023). To AI developers, human annotation is comparably costly and time-intensive.

While human annotation is often considered the gold standard, annotated data still has limits. These stem from unrepresentative annotator pools, incentives that drive down quality, and poor design of annotation interfaces or tasks. Demographic characteristics such as human raters' self-described identities can significantly impact their ratings, raising concerns about introducing biases into annotated datasets (e.g. Aroyo et al. (2023a); Goyal et al. (2022); Homan et al. (2023)). This is especially problematic considering that the demographics of selected human annotators to fine-tune or evaluate generative AI models are rarely representative of the broader population (see, for example, Bai et al. (2022); Ouyang et al. (2022)). To signpost such limitations, it is advisable to analyse and disclose data annotator demographics to indicate whose perspectives have been included in a dataset, as a basic proxy for what biases are likely to exist in that dataset (Aroyo et al., 2023b; Prabhakaran et al., 2021; Sap et al., 2022b).

Concerns have also emerged with respect to aggregating schemes commonly used to compile diverse labels from human annotators into a single ground truth label. These practices have been criticised for leading to an overestimation of future model performance (Gordon et al., 2021), while bypassing important views or disagreements from minority raters on sensitive topics (Field et al., 2021; Marchiori Manerba et al., 2022). Gathering human annotations from online crowdsourcing platforms such as Amazon Mechanical Turk (MTurk), Scale, or Surge is common for large-scale annotation tasks. However, crowdsourcing can be associated with issues of annotation quality, as raters are financially incentivised to bypass certain tasks (e.g. skipping questions) or complete them as fast as possible (Organisciak et al., 2012). Data quality deteriorates as human annotators themselves use AI systems to assist the data annotation work (Veselovsky et al., 2023). Possible ways to mitigate these concerns include equitable pay, payment by time not by task, attention checks, and annotation quality thresholds (Abbey and Meloy, 2017; Jindal, 2022). Careful design of annotation tasks leveraging expertise from psychology experiment set-ups can augment data quality.

[11] Exceptions include harms where reliable algorithmic measures exist, such as privacy harms which can be measured via "memorisation", and debates are ongoing as to the quality of human annotation on factuality (Hosking et al., 2023). Human annotation is also often considered in conjunction with other, basic metrics such as equalised odds to measure fairness.

Benchmarking

The most common evaluation method of model capabilities is benchmarking, which assesses AI system performance against a predefined task, such as mapping AI system outputs to a dataset of prompts and responses. Benchmarks are "success tests", meaning that each task has a clear intended outcome against which model performance is measured. They differ from exploratory tests, where model patterns or "signatures" are the focus (Taylor et al., 2022). Benchmarks may explicitly target failure cases that were identified in a given AI system, thus constituting bespoke challenge datasets. In addition to assessing model output, benchmarks can also be used to assess technical performance metrics, such as energy use at inference (e.g. Kaack et al. (2022); Wang et al. (2020)). Benchmarks have been applied to indicate a range of ethical and social risks – for example, by measuring the likelihood that an AI system outputs toxic or discriminatory content (Cho et al., 2023; Gehman et al., 2020). Benchmarks may be generated from previously annotated data (see Human annotation) (Svikhnushina and Pu, 2023).

Benchmarks ensure test-retest reliability[12] and while they incur computational and financial cost (Liang et al., 2022; Perlitz et al., 2023), they can be comparably cost-effective and time-effective for AI developing organisations,[13] making them suitable for frequent use. Regular testing of an AI system against benchmarks permits tracking model progress over the course of training and allows for cross-model comparisons where metrics are shared in the field. AI developers may run benchmarks regularly over the course of model development to obtain a frequent signal of AI system capabilities that guides AI design ("hill-climbing"). Benchmarks are also used to inform responsible decision-making. Sometimes, evaluations that were initially created for the purposes of responsible decision-making come to be used for "hill-climbing" to bake in ethical and sociotechnical considerations from the beginning; here it is important to keep track of which aim an evaluation is intended to perform and whether it is "held out" or may fall prey to Goodhart's Law as discussed below (see Roles and responsibilities) (Bolukbasi et al., 2016; Gonen and Goldberg, 2019; Liao and Xiao, 2023).

With increasingly anthropomorphic design in generative AI systems and the development of synthetic relatable characters and simulacra (Griffith, 2023; Park et al., 2023a; Pentina et al., 2023), it is tempting to evaluate these systems through methods that were originally devised to assess human or animal psychology (Binz and Schulz, 2023; Bubeck et al., 2023; Frank, 2023). The transfer of cognitive tests to the evaluation of AI has a long history that precedes generative AI systems (e.g. Crosby et al. (2019); Kosoy et al. (2020)). Such psychology-inspired experimentation and benchmarking has been applied to assess whether generative AI systems display certain cognitive capacities, such as Theory of Mind (Sap et al., 2022a) or are "cooperative" (Chan et al., 2023a). However, it is questionable whether applying tests to study constructs such as "empathy" in humans yield any valid or meaningful results when applied to AI systems that are so fundamentally different from human minds (Shiffrin and Mitchell, 2023; Ullman, 2023). Tests that were developed for studying animal cognition or the human mind rely on a range of assumptions (e.g. regarding life cycles, ballpark estimates of memory and learning capacities, and embodiment) which may not hold for AI systems (Mitchell, 2023; Narayanan and Kapoor, 2023). Another problem is that established benchmarks including such experiments suffer validity problems due to "memorisation", where the correct answers may have inadvertently been learned from textual descriptions in AI assistant training data (de Wynter et al., 2023; Mitchell, 2023; Schaeffer et al., 2023).

[12]Test-retest reliability means that running the same benchmark twice should, in theory, lead to the same results. It is undermined where a model has been silently iterated on or fine-tuned in between measurements (Chen et al., 2023; Tu et al., 2023).

[13]Automated benchmarks require technological expertise to set up and incur computational cost, though these are often readily available to large organisations developing generative AI systems. Evaluating smaller versions of an AI system can provide some indication of potential capabilities and risks, but is insufficient as risks may also emerge or become more salient with model size Wei et al. (2022).